

GRAPH DIAMETER

① MILGRAM'S experiment: $d(u, v)$ = distance between nodes u and v

$$\text{avg. distance} = \frac{\sum_{\substack{u, v \in V \\ u \neq v}} d(u, v)}{\binom{|V|}{2}}$$

↳ What if $\max_{\substack{u, v \in V \\ u \neq v}} d(u, v)$? ← DIAMETER

② FINE-GRAINED COMPLEXITY

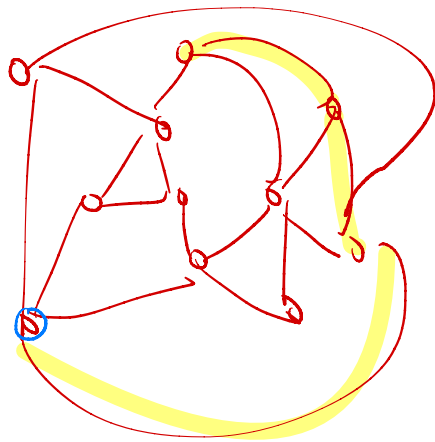
How hard is improving a poly bound?

Graph $G = (V, E)$

$$\begin{aligned} \Rightarrow \text{DIAMETER } D &= \max_{\substack{u, v \in V \\ u \neq v}} d(u, v) \\ &= \max_{u \in V} e(u) \end{aligned}$$

$$\text{where } e(u) = \max_{\substack{v \in V \\ v \neq u}} d(u, v)$$

$D = 3$



We can compute $e(u)$ in $O(|V| + |E|)$ in unweighted graphs using BFS(u) and in $O((|V| + |E|) \log |V|)$ time in weighted graphs using Dijkstra's alg. (no negative cycles)

$\Rightarrow$ TEXTBOOK SOLUTION is $O(nm)$ time, where $n = |V|$, $m = |E|$ and G unweighted

BASELINE ALGORITHM:

Run BFS(u) / Dijkstra(u) for each node $u \in V$

Real-world networks (we saw Fb)

- sparse $m = O(n)$

- huge

$\Rightarrow$ BASELINE

take $O(nm) = O(n^2)$

Issue: real-data networks, the BASELINE is not feasible to run!

What we know as of today:

$$n = |V|$$

- ① No algorithm exists taking $O(n^{2-\epsilon})$ time for any $\epsilon > 0$ unless SETH is false

CONDITIONAL LOWER BOUND

well known
conjecture

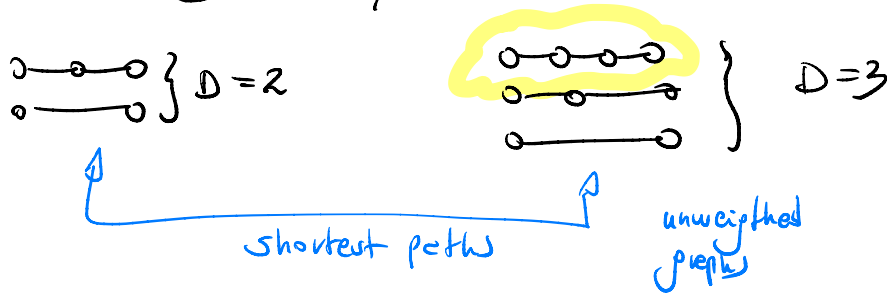
↳ FINE-GRAINED COMPLEXITY

Problems in P $\rightarrow$ poly time but they cannot be improved
unless some well known conjecture is made false.

- ② Good heuristics that are almost linear-time on real-world networks
(no guarantee that it is always so)

- ③ Subquadratic time $O(n \sqrt{n} \log n)$ with a randomized algorithm
that provides $\tilde{D}$ s.t. $\frac{2}{3}D \leq \tilde{D} \leq D$ w.h.p

We focus on ①. In particular, we show that it is already difficult to establish whether $D=2$ or $D=3$ in an arbitrary graph in $O(n^{2-\epsilon})$ time, $\epsilon > 0$



SETH = Strong Exponential-Time Hypothesis

Boolean Formulas: n Boolean variables $x_1, x_2, \dots, x_n \in \{0, 1\}$
F T

literals: x_i or $\bar{x}_i$

clauses: OR of literals: $(x_2 \vee \bar{x}_3 \vee x_5 \vee x_9 \vee \bar{x}_{11})$

formula: AND of clauses: $(x_1 \vee \bar{x}_3) \wedge (x_2 \vee x_3 \vee \bar{x}_5)$

CNF = conjunctive normal form (formula)

assignment: $\tau: x_1, x_2, \dots, x_n \rightarrow \{0, 1\}^n$

τ satisfies a formula $F(x_1, x_2, \dots, x_n)$ iff $F(\tau(x_1), \tau(x_2), \dots, \tau(x_n)) = 1$

EXAMPLE: $F = (x_1 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2 \vee x_3)$

$\tau: x_1, x_2, x_3 \rightarrow 101$

$F(\tau(\cdot)) = (1 \vee \bar{0}) \wedge (\bar{1} \vee 0 \vee 1) = 1 \wedge 1 = 1$

SAT = satisfiability of Boolean formula

Given any formula F , decide if $\exists \tau$ assignment that satisfies F

SAT is NP-complete

BASLINE for SAT is: - generate all $2^{n'}$ possible τ

- check if τ satisfies F in $O(|F|)$ time

$O(2^{n'} \cdot \text{poly}(n')) = \tilde{O}(2^{n'})$ which ignores $\text{poly}(n')$ factors

SETH: No algorithm can solve SAT in $\tilde{O}(2^{\delta n'})$ time for $0 < \delta < 1$
unless $P = NP$

Let's see how to connect diameter 2-vs-3 and SETH

$$G$$

$$n = |V|$$



$$F$$

$n' = \# \text{variables} = \text{even number w.l.o.g.}$
 $x_1, x_2, \dots, x_{n'}$

$|F|$ nodes, one for each clause

0
0
0
0
0
0
0



dummy nodes



$2^{n'/2}$ nodes

each node represents

a distinct assignment of $x_1, x_2, \dots, x_{n'/2}$

0
0
0
0
0
0
0

0
0
0

$2^{n'/2}$ nodes

each node represents

a distinct assignment of $x_{n'/2+1}, \dots, x_{n'}$

total number of nodes is

$$2^{n'/2} + |F| + 2^{n'/2} + 2$$

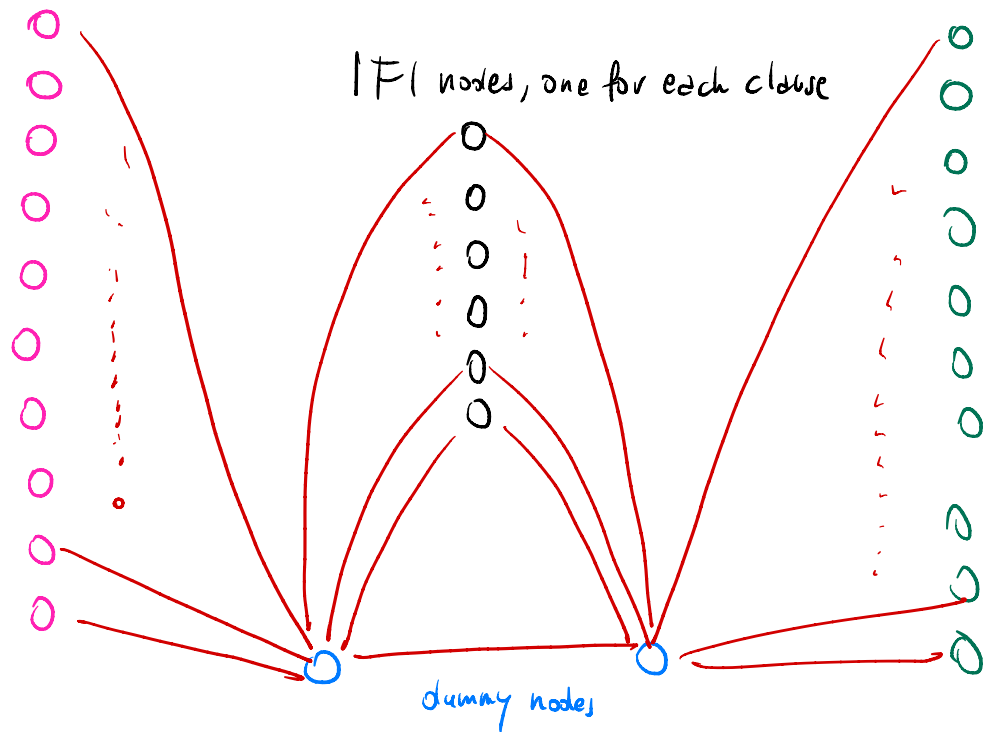
obj if $|F| = O(2^{n'/2})$
 then BASELINE is poly

WLOG, we can assume

$$|F| = O(2^{n'/2})$$

$\Rightarrow$ total number n' of nodes is $O(2^{n'/2})$

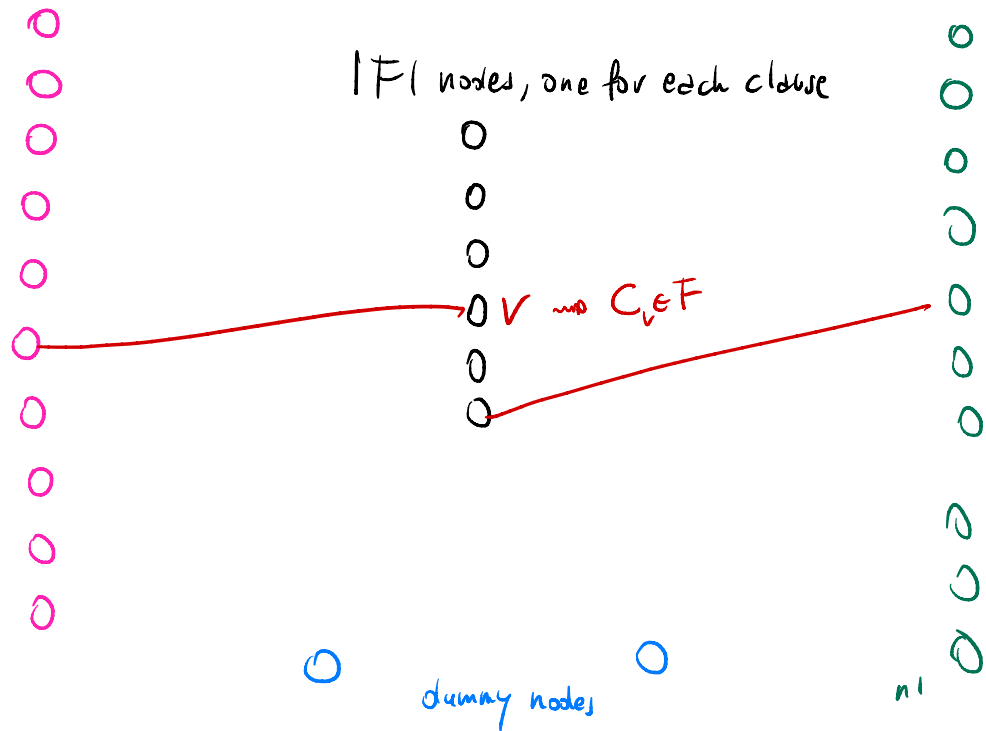
Let's see the edges in this graph: **fixed** + depending on F



Looking at the fixed edges, the diameter is 3:

- ▷ any assignment node on the left is at distance 3 from any assignment node on the right
- ▷ any other pair of nodes is at distance 2

Let's see the edges in this graph: fixed + depending on F



Take any node v in the middle: it represents a clause C_v in the formula F

Consider an assignment node z_i on the left:

We have an edge $(z_i, v) \in E$

iff z_i does not satisfy C_v

Example: $C_v = (x_1 \vee x_2 \vee \bar{x}_3)$

$\rightsquigarrow x_1 \ x_2$

z_i	X	0	1
z_i		0	0

$|E| = O(2^{n/2})$ sparse graph

obs clause C_v is not satisfied by an assignment τ

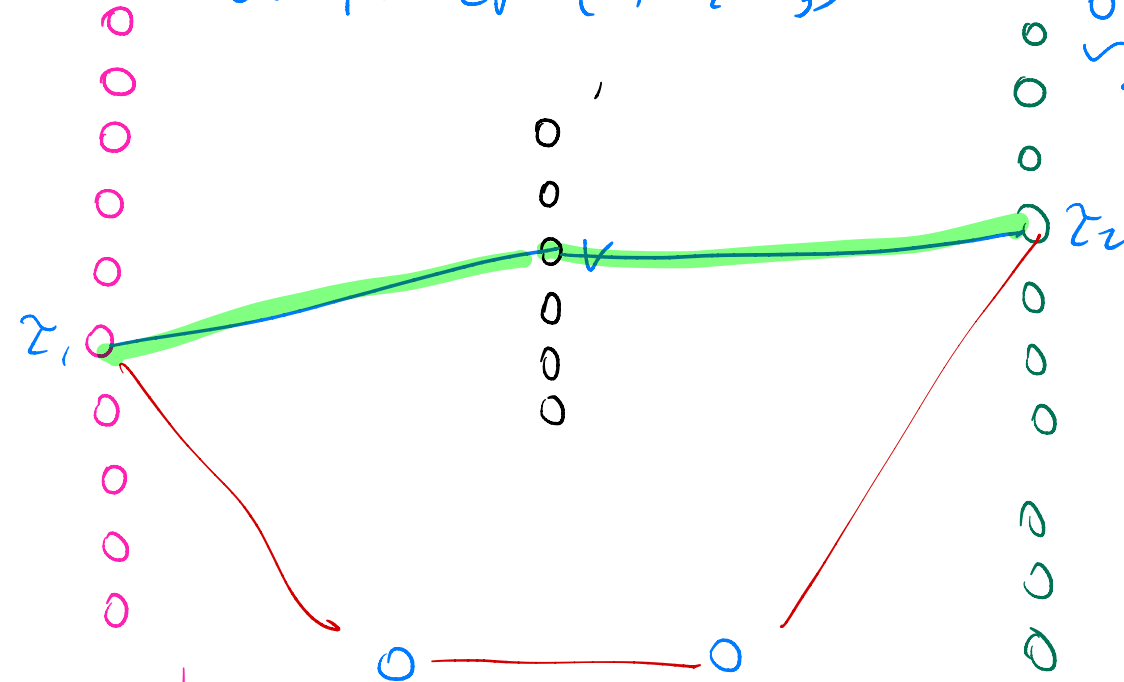
iff $\tau = \tau_1, \tau_2$ and both τ_1 and τ_2 do not satisfy C_v

iff $ol(\tau_1, \tau_2) = 2$

example $C_v = (x_1 \vee x_2 \vee \neg x_3)$

$\tau =$

x_1	x_2	x_3	x_4
0	0	1	-
<u> </u>		<u> </u>	
τ_1		τ_2	



F is not satisfiable iff for each assignment γ , there exists
a clause C_v that is not satisfied

iff there exists γ, γ_2 s.t. both do not
satisfy C_v

iff for each $\gamma, \gamma_2 : d(\gamma, \gamma_2) = 2$

iff $D = 2$

(obv. F is satisfiable iff $D = 3$)
by construction

Summing up: $D = 2 \Rightarrow F$ is not satisfiable
 $D = 3 \Rightarrow F$ is satisfiable

Suppose that there exists an algorithm for $D=2$ vs $D=3$
in $O(n^{2-\epsilon})$ time

Given ^{any} F , we build the previous graph with $N = O(2^{n'/2})$ nodes
with n' variables (and its spouse)

Then we apply the above alg. to decide if $D=2$ or $D=3$

$$\text{Time is } O(n^{2-\epsilon}) = O(\underbrace{(2^{n'/2})}_{n'}^{2-\epsilon}) = O(2^{\underbrace{\frac{2-\epsilon}{2}}_{\delta} n'})$$

$\delta < 1$ as $\epsilon > 0 \Rightarrow$ this gives $O(2^{\delta n'})$ -time alg.

to solve SAT $\Rightarrow$ SETH is false

$\square$